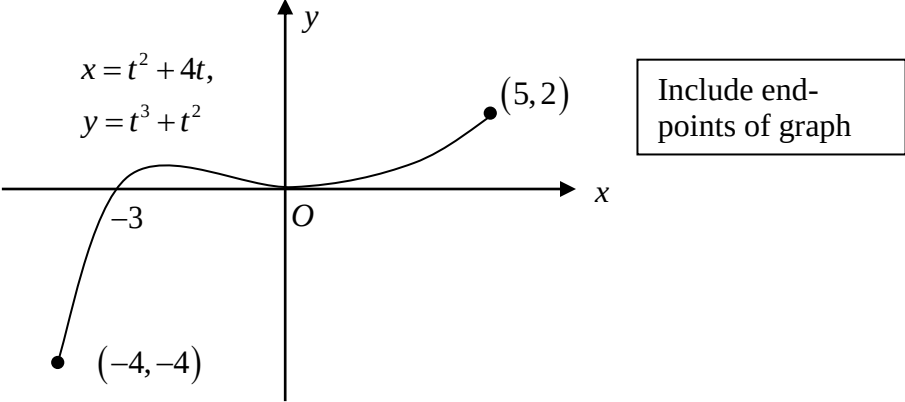
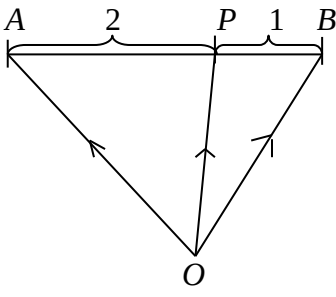


2009 'A' Level H2 Mathematics Paper 2 Suggested Solutions

Section A: Pure Mathematics [40 marks]

Qn	Suggested Solutions
1(i)	
(ii)	$\frac{dx}{dt} = 2t + 4 \quad ; \quad \frac{dy}{dt} = 3t^2 + 2t$ $\Rightarrow \frac{dy}{dx} = \frac{3t^2 + 2t}{2t + 4}$ $\Rightarrow \frac{dy}{dx} \Big _{t=2} = 2$ Equation of the tangent, l i.e. when $t = 2$ $x = 2^2 + 4(2) = 12$, $y = 2^3 + 2^2 = 12$ is: $y - 12 = 2(x - 12)$ $y = 2x - 12$ Alternatively, use parametric mode in the GC, we can obtain the gradient of the curve at $t = 2$.
(iii)	Sub. $x = t^2 + 4t$ and $y = t^3 + t^2$ into l: $t^3 + t^2 = 2(t^2 + 4t) - 12$ $t^3 - t^2 - 8t + 12 = 0$ $(t - 2)(t^2 + t - 6) = 0$ $(t - 2)^2(t + 3) = 0$ $\therefore t = 2$ (rejected) or $t = -3$ when $t = -3$, $x = -3$, $y = -18$ Coordinates of Q: $(-3, -18)$ Note that qn says "Using non-calculator method", do <u>NOT</u> use graphical method here to find the intersection points. Since tangent is at $t = 2$, $(t - 2)$ will be a factor of the equation.

Qn	Suggested Solutions
2(i)	By Ratio Theorem, $\overrightarrow{OP} = \frac{\overrightarrow{OA} + 2\overrightarrow{OB}}{3}$ $= \frac{\begin{pmatrix} 14 \\ 14 \\ 14 \end{pmatrix} + 2 \begin{pmatrix} 11 \\ -13 \\ 2 \end{pmatrix}}{3}$ $= \begin{pmatrix} 12 \\ -4 \\ 6 \end{pmatrix}$ Coordinates of P are $(12, -4, 6)$. 
(ii)	$\overrightarrow{OP} \cdot \overrightarrow{AB} = \begin{pmatrix} 12 \\ -4 \\ 6 \end{pmatrix} \cdot \left[\begin{pmatrix} 11 \\ -13 \\ 2 \end{pmatrix} - \begin{pmatrix} 14 \\ 14 \\ 14 \end{pmatrix} \right]$ $= \begin{pmatrix} 12 \\ -4 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ -27 \\ -12 \end{pmatrix}$ $= 0$ Thus AB and OP are perpendicular.
(iii)	$\mathbf{c} = \frac{1}{\sqrt{12^2 + (-4)^2 + 6^2}} \begin{pmatrix} 12 \\ -4 \\ 6 \end{pmatrix}$ $= \frac{1}{14} \begin{pmatrix} 12 \\ -4 \\ 6 \end{pmatrix}$ $= \begin{pmatrix} \frac{6}{7} \\ -\frac{2}{7} \\ \frac{3}{7} \end{pmatrix}$ $\mathbf{a} \cdot \mathbf{c}$ is the length of projection of $\mathbf{a}$ onto $\mathbf{c}$.

(iv)	$\mathbf{a} \times \mathbf{p} = \begin{pmatrix} 14 \\ 14 \\ 14 \end{pmatrix} \times \begin{pmatrix} 12 \\ -4 \\ 6 \end{pmatrix}$ $= \begin{pmatrix} 140 \\ 84 \\ -224 \end{pmatrix}$ $\mathbf{a} \times \mathbf{p}$ is the area of the parallelogram in which OA and OP are a pair of adjacent sides. Area of triangle $OAP = \frac{1}{2} \mathbf{a} \times \mathbf{p}$ $= \frac{1}{2} \left \begin{pmatrix} 140 \\ 84 \\ -224 \end{pmatrix} \right $ $= \frac{1}{2} \sqrt{76832}$ $= 98\sqrt{2} \text{ square units}$
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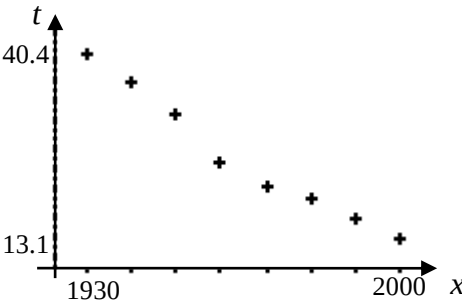
Qn	Suggested Solutions
3(i)	Let $f(x) = y$, $y = \frac{ax}{bx-a}$ $bxy - ay = ax \Rightarrow bxy - ax = ay$ $\therefore x = \frac{ay}{by-a}$ $f^{-1}(x) = \frac{ax}{bx-a}, x \in \mathbb{R}, x \neq \frac{a}{b}$ Since $f(x) = f^{-1}(x)$, $f^2(x) = ff(x) = f^{-1}f(x) = x$. $R_{f^2} = \mathbb{R} \setminus \left\{ \frac{a}{b} \right\}.$ Note: $R_f = \mathbb{R} \setminus \left\{ \frac{a}{b} \right\} = D_f$, that's why f^2 exists.
(ii)	$g(x) = \frac{1}{x}, x \in \mathbb{R}, x \neq 0$ $R_g = \mathbb{R} \setminus \{0\}, D_f = \mathbb{R} \setminus \left\{ \frac{a}{b} \right\}$ Observe that $\left\{ \frac{a}{b} \right\} \in R_g$ but $\left\{ \frac{a}{b} \right\} \notin D_f$.

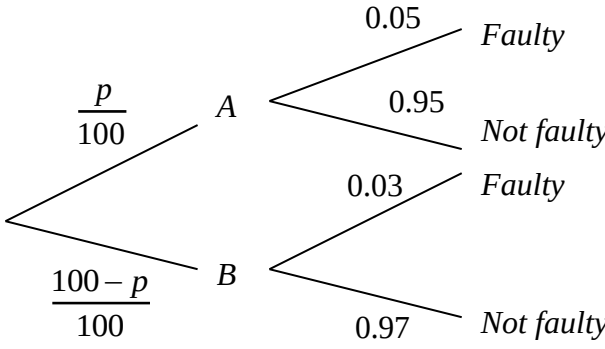
	Since $R_g \not\subseteq D_f$, the composite function fg does not exist.
(iii)	$f^{-1}(x) = x$ $\frac{ax}{bx-a} = x$ $ax = bx^2 - ax$ $bx^2 - 2ax = 0$ $x(bx - 2a) = 0$ $x = 0$ or $x = \frac{2a}{b}$ Do NOT “cancel” x, always do factorisation and conclude from there.

Qn	Suggested Solutions
4(i)	$\frac{d^2n}{dt^2} = 10 - 6t$ $\frac{dn}{dt} = \int 10 - 6t \, dt$ $= 10t - 3t^2 + C$ $n = \int 10t - 3t^2 + C \, dt$ $= 5t^2 - t^3 + Ct + D$ When $t = 0$, $n = 100$ $D = 100$ $\therefore n = 5t^2 - t^3 + Ct + 100$ To sketch family of solution curves (Only for FM students).

(ii)	$\frac{dn}{dt} = 3 - 0.02n$ $\int \frac{1}{3 - 0.02n} dn = \int 1 dt$ $-\frac{1}{0.02} \ln 3 - 0.02n = t + A$ $\ln 3 - 0.02n = -0.02t + B \quad \text{where } B = -0.02A$
Remember the modulus sign	$ 3 - 0.02n = e^{-0.02t + B}$ $3 - 0.02n = Ce^{-0.02t} \quad \text{where } C = \pm e^B$ $0.02n = 3 - Ce^{-0.02t}$ $n = 50(3 - Ce^{-0.02t})$
	When $t = 0$, $n = 100$ $100 = 50(3 - Ce^{-0.02(0)})$ $2 = 3 - C$ $C = 1$ $\therefore n = 50(3 - e^{-0.02t})$ As $t \rightarrow \infty$, $e^{-0.02t} \rightarrow 0$, $n \rightarrow 150$. Therefore, the population will <u>increase</u> and <u>stabilise</u> at (approach the limit of) 150 thousands.

	Section B: Statistics [60 marks]
Qn	Suggested Solutions
5	[Out of Syllabus (9758)]

Qn	Suggested Solutions																
6(i)	 Remember to label: (1) the axes, (2) max and min values on each axis																
(ii)	From the data gathered from 1930 to 2000, a linear model is appropriate from the scatter diagram since most of the data points lie close to a straight line. However, referring to the context of the question, a linear model may not be appropriate in the long run since <u>human capacity will at some point reach a plateau</u> whereby the record cannot be bettered in a linear fashion. Give contextual explanation.																
(iii)	A quadratic model is not suitable for long-term predictions as there will be a point in time where the value of t increases as x increases. This is not an appropriate model since the world record time cannot be higher than before.																
(iv)	In GC, $\ln t = 34.853 - 0.016128x$ $\ln t = 34.9 - 0.0161x \quad (3s.f)$ When $x = 2010$, $\ln t = 34.853 - 0.016128(2010)$ $\ln t = 2.4359$ $t = 11.426$ $= 11.4s$ <table border="1" style="display: inline-table; margin-right: 10px;"> <thead> <tr> <th>L1</th><th>L2</th></tr> </thead> <tbody> <tr><td>1930</td><td>40.4</td></tr> <tr><td>1940</td><td>36.4</td></tr> <tr><td>1950</td><td>31.3</td></tr> <tr><td>1960</td><td>24.5</td></tr> <tr><td>1970</td><td>21.1</td></tr> <tr><td>1980</td><td>19</td></tr> <tr><td>1990</td><td>16.3</td></tr> </tbody> </table> LinReg(a+bx) Xlist: L1 Ylist: L2 FreqList: Store RegEQ: L3 = ln(L2) LinReg y=a+bx a=34.85307066 b=-.0161279508 r²=.9923462737 r=-.9961657863 Thus, the world record time is 3 minutes 41.4 seconds. Even though this is a better model ($r = -0.996$) compared to the linear model ($r = -0.986$), the prediction is still unreliable, as 2010 lies outside the data range and the linear relation between $\ln t$ and x may no longer hold.	L1	L2	1930	40.4	1940	36.4	1950	31.3	1960	24.5	1970	21.1	1980	19	1990	16.3
L1	L2																
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Qn	Suggested Solutions
7(i)	 $P(\text{a randomly chosen component is faulty}) = \frac{25}{100} \times 0.05 + \frac{75}{100} \times 0.03 = \frac{7}{200}$
(ii)	$f(p) = P(\text{component supplied by A} \text{it is faulty})$ $= \frac{\frac{p}{100}(0.05)}{\frac{p}{100}(0.05) + \frac{100-p}{100}(0.03)}$ $= \frac{0.05p}{0.05p + 3 - 0.03p}$ $= \frac{0.05p}{0.02p + 3} \quad (\text{shown})$ Note that it is a conditional probability. $f(p) = \frac{0.05p}{0.02p + 3}$ $f'(p) = \frac{(0.02p + 3)0.05 - 0.05p(0.02)}{(0.02p + 3)^2}$ $= \frac{3}{20(0.02p + 3)^2} > 0$ Therefore, when $0 \leq p \leq 100$, f is an increasing function. Since f is an increasing function, $f(p)$ increases as p increases. Therefore, when the company buys a larger percentage of its electronic components from supplier A, the probability that a randomly chosen component is faulty is supplied by A increases.

Qn	Suggested Solutions
8(i)	Number of ways $= \frac{8!}{3!} = 6720$
(ii)	Number of ways $= \frac{6!}{3!} \times {}^7C_2 \times 2! = 5040$ (Slotting Method) $\begin{array}{ccccccc} & E & L & E & V & A & E \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \end{array}$ Permute the remaining 6 letters, with 3 identical Es Choose 2 slots out of the 7 slots available to slot in T&D. And lastly permute the positions of T&D.
(iii)	Number of ways $= 2 \times 4! \times \frac{4!}{3!} = 192$ 2 possible cases: C V C V C V C V or V C V C V C V C Permute consonants: $4!$ Permute vowels: $\frac{4!}{3!}$ (3 identical E)
(iv)	No of ways Case 1: E _ _ E _ _ E _ $5!$ Case 2: E _ _ E _ _ _ E $5!$ Case 3: E _ _ _ E _ _ E $5!$ Case 4: _ E _ _ E _ _ E $5!$ Number of ways $= 4 \times 5! = 480$

Qn	Suggested Solutions
9(i)	Let M cm be the thickness of a mechanics textbook. $M \sim N(2.5, 0.1^2)$ $\overline{M} \sim N(2.5, \frac{0.1^2}{n})$. $\therefore P(Z > \frac{2.53 - 2.5}{\frac{0.1}{\sqrt{n}}}) = 0.0668$ $\therefore P(Z < \frac{2.53 - 2.5}{\frac{0.1}{\sqrt{n}}}) = 1 - 0.0668 = 0.9332$

	$\frac{2.53 - 2.5}{\frac{0.1}{\sqrt{n}}} = 1.50006$ $\therefore n = 25$ Alternative Method: Using GC, When $n = 25$, $P(\bar{M} > 2.53) = 0.0668$ $\therefore n = 25$ Since n is an integer, we can use the table method. Let $Y_1 = \text{normalcdf}(2.53, \text{E99}, 2.5, 0.1/\sqrt{n})$.
(ii)	Let S cm be the thickness of a statistics textbook. $S \sim N(2.0, 0.08^2)$ Let $T = (M_1 + M_2 + \dots + M_{21}) + (S_1 + S_2 + \dots + S_{24})$ $E(T) = 21 \times 2.5 + 24 \times 2.0 = 100.5$ $\text{Var}(T) = 21 \times 0.1^2 + 24 \times 0.08^2 = 0.3636$ $\therefore T \sim N(100.5, 0.3636)$ Using GC, $P(T < 100) = 0.203$ (3sf) Note that $M_1 + M_2 + \dots + M_{21} \neq 21M$, similarly for statistics textbooks.
(iii)	Let $W = S_1 + S_2 + S_3 + S_4 - 3M$ $E(W) = 4 \times 2.0 - 3 \times 2.5 = 0.5$ $\text{Var}(W) = 4 \times 0.08^2 + 3^2 \times 0.1^2 = 0.1156$ $\therefore W \sim N(0.5, 0.1156)$ Using GC, $P(W < 0) = 0.0707$ (3sf) Remember to square 3 when finding variance of $3M$.
(iv)	We assume that the thicknesses of <u>ALL</u> textbooks are independent of one another.

Qn	Suggested Solutions
10(i)	Unbiased estimate of population mean μ is $\bar{x} = \frac{\sum x}{n} = \frac{86.4}{9} = 9.6$ Unbiased estimate of population variance σ^2 is $s^2 = \frac{n}{n-1} \left(\frac{\sum x^2}{n} - (\bar{x})^2 \right) = \frac{9}{9-1} \left(\frac{835.92}{9} - (9.6)^2 \right) = 0.81$
(ii)	(Only for FM Students) Assumption: Mass of sugar follows a normal distribution. $X \sim N(\mu, \sigma^2) \Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ Let μ denote the population mean mass of sugar the packet.

	$H_0: \mu = 10$ grams $H_1: \mu \neq 10$ grams Test statistic: $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ Level of Significance: 5% Reject H_0 if $p\text{-value} < 0.05$ Under H_0 is true, using GC, $p\text{-value} = 0.21914$ Conclusion: Since $p\text{-value} = 0.21914 > 0.05$, we do not reject H_0 and conclude that there is no sufficient evidence at 5% level, that the mass of the packet is not 10 grams. Explanation: In this case, the sample size is small, it is not large enough to assume a normal distribution according to Central Limit Theorem.
(iii)	(Only for FM Students) As the population variance of X is known, the z-test is carried out instead of the t-test. i.e. $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \therefore Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

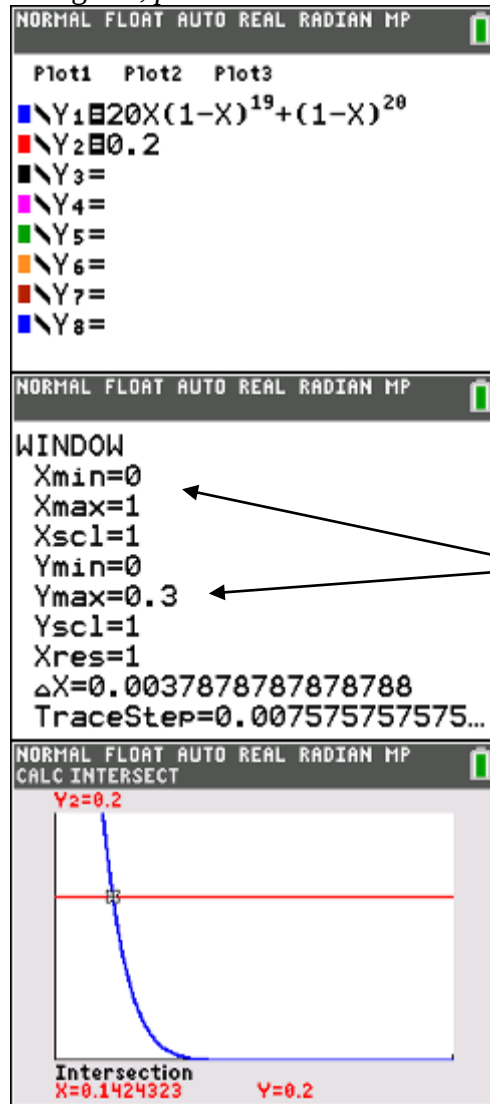
Qn	Suggested Solutions
11(i)	Firstly, the colour of each car must be independent of the colour of any other car. Secondly, the probability of any one car is red is the same throughout the sample.
(ii)	$R \sim B(20, 0.15)$ $P(4 \leq R < 8)$ $= P(4 \leq R \leq 7)$ $= P(R \leq 7) - P(R \leq 3)$ $= 0.34635$ ≈ 0.346
(iii)	(Out of syllabus)
(iv)	(Out of syllabus)

(v) $P(R=1) + P(R=0) = 0.2$

$$\binom{20}{1} p(1-p)^{19} + \binom{20}{0} p^0(1-p)^{20} = 0.2$$

$$20p(1-p)^{19} + (1-p)^{20} = 0.2$$

Using GC, $p = 0.142$



Answer (x-value) should lie between 0 and 1 (because p is probability), and to see intersection at $y = 0.2$, change window setting before graphing.